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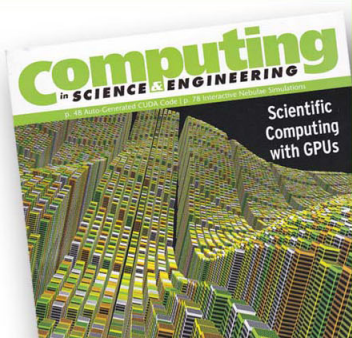
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likely that the method will favor the observation of new spectra that cannot be obtained by other methods, especially in the vacuum ultraviolet region. While there are experimental difficulties in the method, they should be offset by the unique information obtained.

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Molecular-Orbital Theory of Diamagnetism. I. An Approximate LCAO Scheme

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An independent-electron molecular-orbital theory is developed for the diamagnetic behavior of electrons in the presence of an applied magnetic field. The molecular orbitals are written as linear combinations of gauge-invariant atomic orbitals, the dependence of the (complex) coefficients on the magnetic field being studied by perturbation theory. By making a systematic set of approximations involving the neglect of some interatomic terms, a general expression is derived for the diamagnetic susceptibility tensor as a sum of atomic contributions. Each atomic contribution is made up of two parts, the first being a diamagnetic (Langevin-type) term and the second being a paramagnetic contribution involving the details of the electronic excited states. At a lower level of approximation, this second term can be expressed in terms of atomic charge densities and bond orders together with a mean electronic excitation energy.

It is pointed out that the theory provides a convenient basis for detailed interpretation of the Pascal constants used for empirical calculations of diamagnetic susceptibilities of large molecules. In addition, a similar treatment of the theory of nuclear magnetic screening leads to an expression for the shielding constant involving a set of localized contributions closely related to corresponding contributions for the susceptibility.

I. INTRODUCTION

THE diamagnetism of molecules without unpaired electrons has been the subject of a large number of empirical studies.¹ The best-known scheme for interpreting diamagnetic susceptibilities on an atomic or group basis is probably the set of Pascal constants,^{2,3} which permit the calculation of a molar susceptibility on an additive basis (a given contribution for each species of atom) provided that appropriate constitutive corrections are made. In hydrocarbons, for example, the basic constants are adjusted to give reasonable agreement for paraffins and then additional constitutive corrections are proposed for double bonds, triple bonds, phenyl groups and so forth. This scheme has proved successful in coordinating a large body of experimental data within a simple framework.

On the theoretical side, there have been comparatively few studies of molecular diamagnetism for large molecules (other than aromatic hydrocarbons). When the Pascal rules were first formulated, little was known about the fundamental theory, so it was quite

impossible to give the scheme any firm theoretical foundation. However, the development of quantum mechanics opened up the possibility of making *a priori* calculations of molecular properties and Van Vleck⁴ was able to derive a formula which should in principle give the diamagnetic susceptibility in terms of the wave functions for the molecule in the absence of a magnetic field. Owing to the complexity of the calculations, however, detailed applications to other than the simplest molecules have been rare.

Since quantum mechanical methods were first introduced, a great deal of background has been accumulated on electronic structure at an approximate level. In view of the empirical regularity shown by the experimental data on diamagnetism, it should now be possible to develop the basic theory at a level of approximation which will lead to a direct interpretation of the type of constants used in the Pascal scheme. By opening up the possibility of approximate calculations of the constitutive corrections, we might hope to understand their nature better. One of the aims of this and the following papers is to attempt such a theory.

The diamagnetic susceptibility of a molecule is really a tensor quantity with three principal components, the measured value in a gas or liquid being the mean of these. Except for molecules with high symmetry, there

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¹ P. W. Selwood, *Magnetochemistry* (Interscience Publishers, Inc., New York, 1943).

² P. Pascal, *Ann. chim. et phys.* **19**, 5 (1910); **25**, 289 (1912); **29**, 218 (1913).

³ A. Pacault, *Rev. sci.* **86**, 38 (1948).

⁴ J. H. Van Vleck, *Theory of Electric and Magnetic Susceptibilities* (Oxford University Press, New York, 1932).

will be a diamagnetic anisotropy, which can be measured directly in some molecular crystals¹ or by means of the Cotton-Mouton effect⁵ (magnetic double refraction). However, these techniques have not been widely used in recent years and there are comparatively few experimental values of anisotropies available. A comprehensive theory of diamagnetic polarization will, of course, also predict anisotropies and may well suggest some correlations with constitutive corrections that would be worth studying experimentally.

A third reason for theoretical interest in diamagnetism is the connection with chemical shifts in nuclear magnetic resonance. These two phenomena are closely related. The diamagnetic susceptibility of a molecule is equal to the magnetic (dipole) moment of the electronic currents induced by unit (uniform) external magnetic field. These currents produce a secondary magnetic field which has to be added to the primary field to give the total acting on any particular nucleus. The chemical shift of the resonance frequency (for given applied external field) in NMR experiments is therefore determined directly by the same currents as give rise to the susceptibility. As a result, diamagnetic features of particular molecular groups have been used in some theories of chemical shifts.⁶ Ideally there should be a unified theory of diamagnetism giving a joint interpretation of susceptibilities and chemical shifts as different aspects of the same general phenomenon.

The general Van Vleck expression for the molar diamagnetic susceptibility χ_{zz} of a molecule consists of two parts⁴

$$\chi_{zz} = - (Ne^2/4mc^2) \sum_i \langle 0 | x_i^2 + y_i^2 | 0 \rangle + (Ne^2/2m^2c^2) \sum_n' [\langle 0 | L_z | n \rangle]^2 / (E_n - E_0), \quad (1.1)$$

where (x_i, y_i, z_i) are the coordinates of the i th electron, L_z is the z -component total electronic angular momentum operator, and the summation $\sum'$ is over all excited states. The reason why direct computations based on this formula have proved difficult is that the two terms become separately large and nearly cancel in all but the smallest molecules. Similar limitations apply to a direct application of a variational approach.⁷ The difficulties of such a direct approach either to the theory of diamagnetism or chemical shifts lies in the need to select an origin for the vector potential of the uniform field. If this origin is distant from any particular atom, as it must be for a large molecule, two nearly canceling terms inevitably result. A method of avoiding such troubles was first suggested by London⁸ in a theory of the diamagnetism of aromatic hydrocarbons. By choosing molecular orbitals as linear combinations

of atomic orbitals (LCAO) which were themselves modified by a gauge factor (gauge-invariant atomic orbitals), he was able to develop a satisfactory theory of the susceptibility contribution due to π -electron ring currents. Gauge-invariant atomic orbitals have also been used more recently in the theory of the chemical shift^{9,10} and in other calculations of susceptibilities.¹¹

As an LCAO theory approximates each molecular orbital as a sum of atomic contributions, it would seem to be a good starting point for a general theory of atomic contributions to the diamagnetic susceptibility and the chemical shift, provided that appropriate gauge factors are used. In this paper, we shall attempt a theory of diamagnetism starting with simple one-electron LCAO molecular-orbital theory in a magnetic field (that is, a theory in which each electron is treated as moving independently in the same electrostatic potential field). By making a series of rough approximations at a level comparable to those made in corresponding theories without a magnetic field, we shall develop a formula which resembles that used in the Pascal scheme and provides a foundation for approximate calculation of the constants. The corresponding theory of chemical shifts will break down into atomic terms in a similar manner.

II. INDEPENDENT-ELECTRON THEORY OF DIAMAGNETIC POLARIZATION

Following the general approximations made in an independent-electron molecular-orbital theory, the motion of each electron is described by a Hamiltonian operator of the form

$$\mathcal{H} = (2m)^{-1} [\mathbf{p} + (e/c)\mathbf{A}]^2 + V, \quad (2.1)$$

where V is the electrostatic potential of the nuclei together with an averaged screening potential due to other electrons, and $\mathbf{A}$ is the magnetic vector potential. For an electron in the presence of a uniform magnetic field $\mathbf{H}$ and the field of fixed nuclear magnetic dipole moments $\mathbf{u}_1, \mathbf{u}_2, \dots$ at points $\mathbf{R}_1, \mathbf{R}_2, \dots$, the vector potential at a point with position vector $\mathbf{r}$ can be written

$$\mathbf{A}(\mathbf{r}) = \frac{1}{2}(\mathbf{H} \times \mathbf{r}) + \sum_A [\mathbf{u}_A \times (\mathbf{r} - \mathbf{R}_A)] / |\mathbf{r} - \mathbf{R}_A|^3. \quad (2.2)$$

In a theory of diamagnetic susceptibilities, only the first term in this expression need be retained. We restrict ourselves to this case until Sec. V, when the corresponding theory for nuclear magnetic screening will be developed.

In the absence of a magnetic field, LCAO theory approximates solutions of the Schrödinger wave equation by a linear combination of atomic orbitals ϕ_μ , conveniently taken to be real. When there is a magnetic field, however, this is no longer satisfactory as such a

⁵ J. W. Beams, *Revs. Modern Phys.* **4**, 133 (1932).

⁶ J. A. Pople, W. G. Schneider, and H. J. Bernstein *High Resolution Nuclear Magnetic Resonance* (McGraw-Hill Book Company, Inc., New York, 1959).

⁷ J. Tillieu and J. Guy, *Compt. rend.* **240**, 1402 (1955).

⁸ F. London, *J. Phys. Radium* **8**, 397 (1937).

⁹ J. A. Pople, *Proc. Roy. Soc. (London)* **A239**, 541, 550 (1957).

¹⁰ H. F. Hamerka, *Mol. Phys.* **1**, 203 (1958).

¹¹ H. F. Hamerka, *J. Chem. Phys.* **34**, 366, 1996 (1961).

theory does not give a correct description of an isolated atom. Instead, following London,⁸ it is necessary to use gauge-invariant atomic orbitals defined by

$$\chi_\mu = \phi_\mu \exp[-(ie/\hbar c) \mathbf{A}_\mu \cdot \mathbf{r}], \quad (2.3)$$

when $\mathbf{A}_\mu$ is the value of the vector potential $\mathbf{A}$ at the nuclear position $\mathbf{R}_\mu$ of the atomic orbital ϕ_μ . In the presence of a magnetic field, therefore, we seek approximate molecular orbitals ψ of the form

$$\psi_i = \sum_\mu c_{i\mu} \chi_\mu, \quad (2.4)$$

where italic suffixes are used for molecular orbitals and greek ones for atomic orbitals. Application of the variational principle leads to the well-known linear equations

$$\sum_\nu \mathcal{H}_{\mu\nu} c_{i\nu} = \epsilon_i \sum_\nu S_{\mu\nu} c_{i\nu}, \quad (2.5)$$

where ϵ_i is the i th root of the secular equation

$$|\mathcal{H}_{\mu\nu} - \epsilon S_{\mu\nu}| = 0, \quad (2.6)$$

and is the energy of an electron in the molecular orbital ψ_i . The matrix elements $\mathcal{H}_{\mu\nu}$ and $S_{\mu\nu}$ are given by

$$\begin{aligned} \mathcal{H}_{\mu\nu} &= \int \chi_\mu^* \mathcal{H} \chi_\nu d\tau \\ &= \int \exp\{(ie/\hbar c) (\mathbf{A}_\mu - \mathbf{A}_\nu) \cdot \mathbf{r}\} \phi_\mu^* \\ &\quad \cdot [(2m)^{-1} \{\mathbf{p} + (e/c) (\mathbf{A} - \mathbf{A}_\nu)\}^2 + V] \phi_\nu d\tau, \end{aligned} \quad (2.7)$$

and

$$\begin{aligned} S_{\mu\nu} &= \int \chi_\mu^* \chi_\nu d\tau \\ &= \int \exp\{(ie/\hbar c) (\mathbf{A}_\mu - \mathbf{A}_\nu) \cdot \mathbf{r}\} \phi_\mu^* \phi_\nu d\tau. \end{aligned} \quad (2.8)$$

It should be noted that because of the gauge-factors, the overlap integrals depend on the magnetic field as well as the Hamiltonian integrals $\mathcal{H}_{\mu\nu}$.

To develop a theory of diamagnetic susceptibilities from these equations, we shall make a series of systematic approximations, the effect of which will be considered successively.

Approximation A. As suggested by London,⁸ the exponential factors appearing in (2.7) and (2.8) will be taken outside the integral sign, replacing $\mathbf{r}$ by $\frac{1}{2}(\mathbf{R}_\mu + \mathbf{R}_\nu)$.

This approximation is not required if ϕ_μ and ϕ_ν are atomic orbitals on the same atom. If they are on different atoms, the approximation is reasonable, since the midpoint between nuclei μ and ν will be close to the centre of the "overlap distribution" $\phi_\mu \phi_\nu$. One drawback of the approximation is that it may make the matrix $\mathcal{H}_{\mu\nu}$ non-Hermitian. However, this is easily put

right by splitting the resulting matrix into Hermitian and anti-Hermitian parts and retaining only the former. If we write

$$L_{\mu\nu} = (e/2\hbar c) (\mathbf{A}_\mu - \mathbf{A}_\nu) \cdot (\mathbf{R}_\mu + \mathbf{R}_\nu), \quad (2.9)$$

the approximate matrix elements can then be written

$$\begin{aligned} 2\mathcal{H}_{\mu\nu} &= \exp(iL_{\mu\nu}) \int \phi_\mu^* [(2m)^{-1} \{\mathbf{p} + (e/c) (\mathbf{A} - \mathbf{A}_\nu)\}^2 + V] \\ &\quad \cdot \phi_\nu d\tau + \exp(iL_{\nu\mu}) \\ &\quad \cdot \int \phi_\nu^* [(2m)^{-1} \{\mathbf{p} + (e/c) (\mathbf{A} - \mathbf{A}_\mu)\}^2 + V] \phi_\mu d\tau, \end{aligned} \quad (2.10)$$

$$S_{\mu\nu} = \exp(iL_{\mu\nu}) \int \phi_\mu^* \phi_\nu d\tau. \quad (2.11)$$

Approximation B. It is now assumed that the magnetic field is weak enough for all magnetic energies to be small compared with the energy differences between molecular orbitals $E_i - E_j$.

This is unexceptionable as long as we are not dealing with degenerate molecular orbitals. It will become clear later that the theory applies even if degenerate orbitals are present, as long as they are completely full.

As a result of approximation B, magnetic effects can be treated as perturbations and all matrix elements can be expanded in orders of magnitude in $\mathbf{A}$,

$$\begin{aligned} \mathcal{H}_{\mu\nu} &= \mathcal{H}_{\mu\nu}^{(0)} + \mathcal{H}_{\mu\nu}^{(1)} + \mathcal{H}_{\mu\nu}^{(2)} + \dots, \\ S_{\mu\nu} &= S_{\mu\nu}^{(0)} + S_{\mu\nu}^{(1)} + S_{\mu\nu}^{(2)} + \dots \end{aligned} \quad (2.12)$$

Explicitly

$$\begin{aligned} \mathcal{H}_{\mu\nu}^{(0)} &= \int \phi_\mu^* \mathcal{H}^{(0)} \phi_\nu d\tau, \\ H_{\mu\nu}^{(0)} &= iL_{\mu\nu} \mathcal{H}_{\mu\nu}^{(0)} + (e/mc) \left[\int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu) \cdot \mathbf{p} \phi_\nu d\tau \right]_{\text{Herm}}, \\ \mathcal{H}_{\mu\nu}^{(2)} &= -\frac{1}{2} L_{\mu\nu}^2 \mathcal{H}_{\mu\nu}^{(0)} + (ie/mc) L_{\mu\nu} \\ &\quad \cdot \left[\int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu) \cdot \mathbf{p} d\tau \right]_{\text{Herm}} + (e^2/2mc^2) \\ &\quad \cdot \left[\int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu)^2 \phi_\nu d\tau \right]_{\text{Herm}}, \end{aligned} \quad (2.13)$$

where $\mathcal{H}^{(0)}$ is the Hamiltonian in the absence of a magnetic field. We have adopted the general matrix notation for the Hermitean part

$$[B_{\mu\nu}]_{\text{Herm}} = \frac{1}{2} (B_{\mu\nu} + B_{\nu\mu}^*). \quad (2.14)$$

A similar expansion can be made for the overlap matrix element $S_{\mu\nu}$.

We now make the following further approximations.

Approximation C. It is assumed that atomic orbitals ϕ_μ on the same atom are mutually or-

thogonal and further that the overlap integral between atomic orbitals on different atoms may be neglected.

While the second part of this approximation is not necessarily quantitatively accurate, it leads to major simplifications and is widely made in molecular orbital theories of nonmagnetic phenomena. As a result the overlap matrix takes the form

$$S_{\mu\nu} = \delta_{\mu\nu}, \quad (2.15)$$

and account no longer has to be taken of its dependence on magnetic field. We also make two further related approximations to simplify the matrix elements $\mathcal{H}^{(1)}$ and $\mathcal{H}^{(2)}$

Approximation D. The integral $\int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu) \cdot \mathbf{p} \phi_\nu d\tau$ is neglected if ϕ_μ and ϕ_ν are on different atoms.

The justification of this approximation is as follows. For a uniform external field, the operator $(\mathbf{A} - \mathbf{A}_\nu) \cdot \mathbf{p}$ is proportional to the orbital angular momentum operator centred on nucleus ν . If ϕ_ν is an atomic s orbital, it is annihilated by the operator and the integral vanishes. If it is a p or d orbital, the operator converts it into another p or d orbital *on the same atom*, and we are left with an overlap-type integral between different atoms which is then neglected.

Approximation E. The integral $\int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu)^2 \phi_\nu d\tau$ is neglected unless ϕ_μ and ϕ_ν are identical.

Approximations C, D, and E are consistent with one another in that they neglect the magnitude of the overlap charge and certain weighted distributions of it. Only the elements $\mathcal{H}_{\mu\nu}^{(0)}$ are retained between different atoms. The approximations are closely related to those often used in semiempirical theories of conjugated hydrocarbons, often referred to as the neglect of differential overlap.

As a result of approximation D, the second term in $\mathcal{H}_{\mu\nu}^{(2)}$ always drops out as $L_{\mu\nu}$ vanishes if the integral does not. The need for taking the Hermitean part also disappears. The remaining expressions are conveniently expressed using the symbol

$$\eta_{\mu\nu} = \begin{cases} 1 & \text{if } \phi_\mu \text{ and } \phi_\nu \text{ are on the same atom,} \\ 0 & \text{otherwise.} \end{cases} \quad (2.16)$$

Then

$$\begin{aligned} \mathcal{H}_{\mu\nu}^{(1)} &= i L_{\mu\nu} \mathcal{H}_{\mu\nu}^{(0)} + (e/mc) \eta_{\mu\nu} \int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu) \cdot \mathbf{p} \phi_\nu d\tau, \\ \mathcal{H}_{\mu\nu}^{(2)} &= -\frac{1}{2} L_{\mu\nu}^2 \mathcal{H}_{\mu\nu}^{(0)} + (e^2/2mc^2) \delta_{\mu\nu} \int \phi_\mu^* (\mathbf{A} - \mathbf{A}_\nu)^2 \phi_\nu d\tau. \end{aligned} \quad (2.17)$$

In both these expressions, the first term is *interatomic* and the second is *intraatomic*. The former are the basis

of the London theory of ring currents in aromatic systems; the latter will describe circulations of an atomic nature.

To calculate the molar diamagnetic susceptibility tensor, we have to find the change in total energy to second order in the applied field H . If $\alpha, \beta, \dots$ are tensor suffixes, the susceptibility tensor is given by

$$E^{(2)} = -\frac{1}{2} N^{-1} \chi_{\alpha\beta} H_\alpha H_\beta, \quad (2.18)$$

where a summation is implied by repeated suffixes. We therefore need the corresponding second order energy change for each orbital energy ϵ_i . If $c_{i\mu}^{(0)}$ are the LCAO coefficients in the unperturbed problem (at the same level of approximation), standard perturbation theory gives, in the absence of degeneracy,

$$\begin{aligned} \epsilon_i^{(2)} &= \sum_{\mu\nu} c_{i\mu}^{(0)*} \mathcal{H}_{\mu\nu}^{(2)} c_{i\nu}^{(0)} - \sum_{j(\neq i)} \sum_{\mu\lambda\nu\sigma} c_{i\mu}^{(0)*} c_{j\lambda}^{(0)*} \\ &\quad \cdot [\mathcal{H}_{\mu\nu}^{(1)} \mathcal{H}_{\lambda\sigma}^{(1)} / (\epsilon_j^{(0)} - \epsilon_i^{(0)})] c_{j\nu}^{(0)} c_{i\sigma}^{(0)}. \end{aligned} \quad (2.19)$$

Here $\sum_{j(\neq i)}$ is a sum over *all* the unperturbed molecular orbitals except ψ_i in this LCAO framework. (The total number of molecular orbitals will be equal to the total number of atomic orbitals in the basis considered.) To obtain the total susceptibility we sum twice over each *occupied* molecular orbital ψ_i (assuming each to be doubly occupied), giving

$$\begin{aligned} E^{(2)} &= \sum_{\mu\nu} P_{\mu\nu} \mathcal{H}_{\mu\nu}^{(2)} - 2 \sum_i^{\text{occ}} \sum_{j(\neq i)}^{\text{unocc}} \sum_{\mu\lambda\nu\sigma} c_{i\mu}^{(0)*} c_{j\lambda}^{(0)*} \\ &\quad \cdot [\mathcal{H}_{\mu\nu}^{(1)} \mathcal{H}_{\lambda\sigma}^{(1)} / (\epsilon_j^{(0)} - \epsilon_i^{(0)})] c_{j\nu}^{(0)} c_{i\sigma}^{(0)}, \end{aligned} \quad (2.20)$$

where we have used a population-density matrix

$$P_{\mu\nu} = 2 \sum_i^{\text{occ}} c_{i\mu}^{(0)*} c_{i\nu}^{(0)}. \quad (2.21)$$

It has been possible to replace the summation $\sum_{j(\neq i)}$ by the summation over unoccupied orbitals $\sum_{j \text{ unocc}}$, since terms in the double sum $\sum_{i \text{ occ}} \sum_{j(\neq i) \text{ occ}}$ cancel in pairs. As pointed out by Coulson and Longuet-Higgins¹² in their development of the corresponding theory of electrostatic perturbations, the form of (2.20) justifies the application of the method to molecules with degenerate molecular orbitals, provided that these are completely full, for no terms become infinite if the energies of two occupied orbitals become identical.

III. SIMPLIFICATION FOR NONCYCLIC MOLECULES

It has already been pointed out that the expressions (2.17) for the first and second order perturbations to the Hamiltonian matrix elements consist of interatomic and intra-atomic parts. The interatomic terms give rise to ring currents flowing round a cyclic set of bonds in a molecule; it might therefore be expected on physical grounds that such effects cannot occur in noncyclic molecules with no such closed loops. In this section, we shall show that this follows mathematically

¹² C. A. Coulson and H. C. Longuet-Higgins, Proc. Roy. Soc. (London) **A191**, 39 (1947).

from the assumptions made and that, for noncyclic molecules, the terms involving $L_{\mu\nu}$ in (2.17) may be omitted in energy calculations.

To demonstrate this, it is convenient to return to the secular equation

$$\begin{aligned}\det(K_{\mu\nu}) &= 0, \\ K_{\mu\nu} &= \mathcal{H}_{\mu\nu} - \epsilon \delta_{\mu\nu},\end{aligned}\quad (3.1)$$

and to note that, to the approximation considered, $\mathcal{H}_{\mu\nu}$ may be written in the form

$$\mathcal{H}_{\mu\nu} = \exp(iL_{\mu\nu})\mathcal{H}_{\mu\nu}^{(0)} + \mathcal{H}_{\mu\nu}^+, \quad (3.2)$$

where $\mathcal{H}_{\mu\nu}^+$ is the intra-atomic part of the corrections (2.17). Now each term in the complete expansion of the determinant (3.1) will consist of a product of elements $K_{\mu\nu}$ in which each suffix μ will appear twice, once as first suffix and once as second. In such a term, a particular suffix μ will appear (i) as a diagonal element $K_{\mu\mu}$, or (ii) as part of a cyclic product $K_{\mu\nu}K_{\nu\sigma}K_{\sigma\rho}\cdots K_{\tau\mu}$, the suffixes $\mu, \nu, \sigma, \rho, \cdots, \tau$ being all different. The diagonal elements do not involve the interatomic terms in any case, so we need only consider the effects of the $\exp(iL_{\mu\nu})$ factors on the cyclic products of type (ii). Here we note that

$$K_{\mu\nu}K_{\nu\sigma}\cdots K_{\tau\mu} = \mathcal{H}_{\mu\nu}\mathcal{H}_{\nu\sigma}\cdots\mathcal{H}_{\tau\mu}. \quad (3.3)$$

All the matrix elements $\mathcal{H}_{\mu\nu}, \mathcal{H}_{\nu\sigma}, \cdots$ in this product are either intra- or interatomic. However, for noncyclic molecules, if a particular one (say $\mathcal{H}_{\nu\sigma}$) is interatomic, ϕ_ν being on atom A and ϕ_σ being on atom B, then there must be another element later in the product (say $\mathcal{H}_{\lambda\pi}$) with ϕ_λ on B and ϕ_π on A. This is because the series $\mu, \nu, \sigma, \cdots, \tau, \mu$ has eventually to return to the original atomic orbital ϕ_μ . It follows that

$$L_{\nu\sigma} = -L_{\lambda\pi}. \quad (3.4)$$

Further, we may note that $\mathcal{H}_{\mu\nu}^+$ vanishes for all interatomic terms at the level of approximation we are using. It follows that

$$\mathcal{H}_{\nu\sigma}\mathcal{H}_{\lambda\pi} = \mathcal{H}_{\nu\sigma}^{(0)}\mathcal{H}_{\lambda\pi}^{(0)}, \quad (3.5)$$

so that the exponential factors appearing in (3.3) annihilate each other in pairs. This means that correct energy levels are obtained if all such factors are omitted in the secular equation, corresponding to the omission of the first term in the perturbation expressions (2.17).

IV. ATOMIC CONTRIBUTIONS TO THE DIAMAGNETIC SUSCEPTIBILITY OF NONCYCLIC MOLECULES

After omitting the interatomic terms, the Hamiltonian matrix element perturbations take the form

$$\begin{aligned}\mathcal{H}_{\mu\nu}^{(1)} &= -(ie\hbar/2mc)\eta_{\mu\nu}H_\alpha \int \phi_\mu^* l_\alpha \phi_\nu d\tau = -(ie\hbar/2mc) \\ &\quad \cdot \eta_{\mu\nu}H_\alpha (l_\alpha)_{\mu\nu}, \\ \mathcal{H}_{\mu\nu}^{(2)} &= (e^2/8mc^2)\delta_{\mu\nu}H_\alpha H_\beta \int \phi_\mu^* (r^2\delta_{\alpha\beta} - r_\alpha r_\beta) \phi_\nu d\tau,\end{aligned}\quad (4.1)$$

where $\mathbf{l}$ is the angular momentum vector $\mathbf{r}\times\nabla$. All of these integrals are one center in character and in each case the origin is chosen at the nucleus of that particular atom.

Substituting into (2.20) and noting that $\mathcal{H}_{\nu\mu}^{(1)} = -\mathcal{H}_{\mu\nu}^{(1)}$, we obtain an expression for the diamagnetic susceptibility in two parts

$$\chi_{\alpha\beta} = (\chi_d)_{\alpha\beta} + (\chi_p)_{\alpha\beta}, \quad (4.2)$$

where

$$(\chi_d)_{\alpha\beta} = -(Ne^2/4mc^2) \sum_\mu P_{\mu\mu} (r^2\delta_{\alpha\beta} - r_\alpha r_\beta)_{\mu\mu} \quad (4.3)$$

and

$$\begin{aligned}(\chi_p)_{\alpha\beta} &= (Ne^2\hbar^2/m^2c^2) \sum_i^{\text{occ}} \sum_j^{\text{unocc}} (\epsilon_j - \epsilon_i)^{-1} \\ &\quad \cdot \sum_{\mu < \nu} \sum_{\lambda < \sigma} \eta_{\mu\nu} \eta_{\lambda\sigma} (c_{i\mu} c_{j\nu} - c_{i\nu} c_{j\mu}) (c_{i\lambda} c_{j\sigma} - c_{i\sigma} c_{j\lambda}) (l_\alpha)_{\mu\nu} (l_\beta)_{\lambda\sigma}.\end{aligned}\quad (4.4)$$

The zero superscript on the LCAO coefficients $c_{i\mu}^{(0)}$ has now been omitted to simplify the notation and all such coefficients are supposed real.

The contribution χ_d to the total susceptibility can be split into a sum of individual contributions from each atom in the molecule

$$\chi_d = \sum_A \chi_d^A,$$

$$(\chi_d^A)_{\alpha\beta} = -(Ne^2/4mc^2) \sum_\mu^A P_{\mu\mu} (r^2\delta_{\alpha\beta} - r_\alpha r_\beta)_{\mu\mu}. \quad (4.5)$$

Where $\sum_\mu^A$ is a sum over all atomic orbitals on atom A. This expression gives just the diamagnetism that would be calculated for a molecule if the Langevin atomic formula were applied to each atomic orbital with the appropriate total electron population $P_{\mu\mu}$. For the z component on atom A, (4.5) takes the form

$$(\chi_d^A)_{zz} = -(Ne^2/4mc^2) \sum_\mu^A P_{\mu\mu} (x^2 + y^2)_{\mu\mu}. \quad (4.6)$$

The second contribution χ_p depends on the occupied and unoccupied molecular orbitals in a rather detailed way and is likely to be more sensitive to the chemical nature of the molecule. However, given adequate knowledge of these orbitals and their energies, (4.3) and (4.4) provide a direct method for calculating the susceptibility. The two parts χ_d and χ_p may be described as diamagnetic and (temperature-independent) paramagnetic contributions in this gauge-invariant LCAO theory, but they are not to be directly compared with the two parts of the original Van Vleck formula, which cannot be divided up into atomic contributions in such a satisfactory manner.

The paramagnetic contribution can also be written

as a sum of atomic terms

$$\chi_p = \sum_A \chi_p^A$$

$$(\chi_p^A)_{\alpha\beta} = (Ne^2\hbar^2/m^2c^2) \sum_i^{\text{occ}} \sum_j^{\text{unocc}} (\epsilon_j - \epsilon_i)^{-1}$$

$$\cdot \sum_{\mu < \nu} \sum_B \sum_{\lambda < \sigma} (c_{i\mu}c_{j\nu} - c_{i\nu}c_{j\mu}) (c_{i\lambda}c_{j\sigma} - c_{i\sigma}c_{j\lambda}) (l_\alpha)_{\mu\nu} (l_\beta)_{\lambda\sigma}. \quad (4.7)$$

This can be interpreted as the paramagnetic moment per unit field induced on atom A. Unlike the diamagnetic part (4.5), it involves a sum $\sum_B$ over all atoms in the molecule, since a magnetic field acting on an atom B will give rise to paramagnetic moments on atom A.

The atomic paramagnetic susceptibility (4.7) can be put in more explicit terms if the types of atomic orbitals involved are considered in detail. If we consider the zz component $(\chi_p^A)_{zz}$, the matrix elements $l_{\mu\nu}$ are then only non-vanishing between pairs of atomic orbitals such as (p_x, p_y) , (d_{zz}, d_{yz}) , $(d_{x^2-y^2}, d_{xy}) \dots$. To avoid too complicated an expression, we shall restrict ourselves in the remainder of this section to molecules in which the LCAO scheme can be developed satisfactorily with only atomic s and p functions. This would apply to compounds of elements in the first row of the periodic table and also many in the second row. With this restriction and noting that $l_{p_x, p_y} = 1$, (4.7) may be written

$$(\chi_p^A)_{zz} = (Ne^2\hbar^2/m^2c^2) \sum_i^{\text{occ}} \sum_j^{\text{unocc}} (\epsilon_j - \epsilon_i)^{-1}$$

$$\cdot (c_{ix_A}c_{jy_A} - c_{iy_A}c_{jx_A}) \sum_B (c_{ix_B}c_{jy_B} - c_{iy_B}c_{jx_B}). \quad (4.8)$$

Here c_{ix_A} is written for the coefficient of the p_x orbital on atom A etc.

An additional simplification can be made if the inverse excitation energy factors $(\epsilon_j - \epsilon_i)^{-1}$ are replaced by a mean value $\langle \Delta E^{-1} \rangle_{Av}$. The summation over i and j can then be expressed in terms of the generalized bond-order matrix [Eq. (2.20)] since

$$\sum_i^{\text{occ}} c_{i\mu}c_{i\nu} + \sum_j^{\text{unocc}} c_{j\mu}c_{j\nu} = \delta_{\mu\nu}, \quad (4.9)$$

(in the approximation of neglect of overlap integrals). We then get

$$(\chi_p^A)_{zz} = (Ne^2\hbar^2/4m^2c^2) \langle (\Delta E)^{-1} \rangle_{Av}$$

$$\cdot \sum_B [P_{xAxB}(2\delta_{AB} - P_{yAyB}) + P_{yAyB}(2\delta_{AB} - P_{xAxB})$$

$$+ 2P_{xAyB}P_{yAxB}]. \quad (4.10)$$

This is a generalization of a formula previously given⁹ for simple hydrides XH_n . Similar formulas can be developed for the other components of the tensor.

These equations now provide the basis for approximate calculations of atomic contributions to the total

susceptibility of a large molecule, provided that sufficient information is available about the molecular orbitals. In a following paper, such calculations will be described for some characteristic organic groups and compared with the empirical Pascal constants.

V. ATOMIC CONTRIBUTIONS TO THE NMR CHEMICAL SHIFTS FOR NONCYCLIC MOLECULES

A theory of nuclear magnetic resonance chemical shifts can be developed at the same level of approximation using the complete vector potential given by Eq. (2.2). For simplicity consider a molecule with only one nuclear moment $\mathbf{u}$ at the nuclear position of atom A, conveniently taken to be the origin. Then

$$\mathbf{A}(\mathbf{r}) = \frac{1}{2}\mathbf{H}\mathbf{x}\mathbf{r} + \mathbf{u}\mathbf{x}\mathbf{r}/r^3. \quad (5.1)$$

A set of approximations similar to those already described in Sec. 2 reduces the matrix elements $\mathcal{H}_{\mu\nu}^{(1)}$ and $\mathcal{H}_{\mu\nu}^{(2)}$ to the forms (2.17) (without the interatomic $L_{\mu\nu}$ terms). These involve only local vector potentials cated by the r^{-3} terms in (5.1) and it is convenient to make one further approximation.

Approximation F. The local vector potential $(\mathbf{A} - \mathbf{A}_v)$ is replaced by the local vector potential of a uniform field equal to that given by (5.1) at the center v . For the atom A with the nuclear moment, the correct form (5.1) is retained. Quantitatively, this gives

$$\mathbf{A} - \mathbf{A}_v = \frac{1}{2}\mathbf{H}\mathbf{x}\mathbf{r}(\mathbf{r} - \mathbf{R}_v), \quad (5.2)$$

$$(H_v)_\alpha = H_\alpha + R_v^{-5} [R_v^2 \delta_{\alpha\beta} - 3(R_v)_\alpha (R_v)_\beta] \mu_\beta. \quad (5.3)$$

It is a valid approximation at large distances and corresponds to treating interatomic shielding effects on a dipole basis. The second order perturbation energy expression (2.18) has to be replaced by

$$E^{(2)} = -N^{-1} \chi_{\alpha\beta} H_\alpha H_\beta + \sigma_{\alpha\beta}^A H_\alpha \mu_\beta, \quad (5.4)$$

where $\sigma_{\alpha\beta}^A$ is the magnetic shielding tensor for atom A.

To find the components of the tensor $\sigma_{\alpha\beta}^A$, we require the new perturbation to the Hamiltonian matrix elements $\mathcal{H}_{\mu\nu}$. Only intra-atomic terms are nonzero and for these we find

(a) ϕ_μ and ϕ_ν both on atom A

$$\mathcal{H}_{\mu\nu}^{(1)} = -(ie\hbar/2mc) [H_\alpha (l_\alpha)_{\mu\nu} + 2\mu_\alpha (\tau^{-3} l_\alpha)_{\mu\nu}], \quad (5.5)$$

$$\mathcal{H}_{\mu\nu}^{(2)} = (e^2/8mc^2) \delta_{\mu\nu}$$

$$\cdot [H_\alpha H_\beta (\tau^2 \delta_{\alpha\beta} - \mathbf{r}_\alpha \mathbf{r}_\beta)_{\mu\nu} + 4H_\alpha \mu_\beta (r^{-1} \delta_{\alpha\beta} - r^{-3} \mathbf{r}_\alpha \mathbf{r}_\beta)_{\mu\nu}]; \quad (5.6)$$

(b) ϕ_μ and ϕ_ν both on atom B ($\neq A$)

$$\mathcal{H}_{\mu\nu}^{(1)} = -(ie\hbar/2mc)$$

$$\cdot [H_\alpha + R_B^{-5} \{R_B^2 \delta_{\alpha\beta} - 3R_{B\alpha} R_{B\beta}\} \mu_\beta] (l_\alpha)_{\mu\nu}, \quad (5.7)$$

$$\mathcal{H}_{\mu\nu}^{(2)} = (e^2/8mc^2) \delta_{\mu\nu}$$

$$\cdot [H_\alpha H_\beta + 2H_\alpha \mu_\beta R_B^{-5} \{R_B^2 \delta_{\beta\gamma} - 3R_{B\beta} R_{B\gamma}\}] (r^2 \delta_{\alpha\gamma} - \mathbf{r}_\alpha \mathbf{r}_\gamma)_{\mu\nu}, \quad (5.8)$$

terms of order μ^2 having been omitted.

Substitution of these elements into the general perturbation formula (2.20) leads to an expression for the second order perturbation energy $E^{(2)}$. Comparison with (5.4) yields $\sigma_{\alpha\beta}^A$. The complete formula for this tensor contains many parts which can be expressed in terms of the atomic contributions to the magnetic susceptibility derived in previous sections of the paper. The final result is

$$\sigma_{\alpha\beta}^A = (e^2/2mc^2) \sum_{\mu}^A P_{\mu\mu} (r^{-1}\delta_{\alpha\beta} - r^{-3}r_{\alpha}r_{\beta})_{\mu\mu} - 2N^{-1} \cdot \langle r^{-3} \rangle_A (\chi_p^A)_{\alpha\beta} - N^{-1} \sum_{B(\neq A)} \chi_{\alpha\gamma}^B R_B^{-5} (R_B^2\delta_{\gamma\beta} - 3R_{B\gamma}R_{B\beta}). \quad (5.9)$$

Here $\langle r^{-3} \rangle_A$ is a mean value of r^{-3} for the atomic orbitals contributing to the paramagnetic terms on atom A. If only one set of p orbitals is involved, no approximation is implied and $\langle r^{-3} \rangle_A$ is just the mean of r^{-3} for those p orbitals.

Equation (5.9) gives the full shielding tensor $\sigma_{\alpha\beta}^A$. In most practical NMR experiments, only the mean value of this tensor is of interest. The corresponding expression is

$$\sigma^A = (e^2/3mc^2) \sum_{\mu}^A P_{\mu\mu} (r^{-1})_{\mu\mu} - 2N^{-1} \langle r^{-3} \rangle_A \chi_p^A - \frac{1}{3}N^{-1} \sum_{B(\neq A)} \chi_{\alpha\gamma}^B R_B^{-5} (R_B^2\delta_{\gamma\alpha} - 3R_{B\gamma}R_{B\alpha}). \quad (5.10)$$

The three parts of these expressions for the shielding tensor and its mean value are easily interpreted. The first reduces to the Lamb formula¹³ for the atom A. For an isolated atom in a spherically symmetric ground

state, this is the only nonvanishing term and gives the total screening. The second term gives the paramagnetic contribution of the atomic orbitals on atom A. The contribution to σ is negative. For hydrogen atoms this term vanishes, insofar as $2p$ atomic orbitals do not have to be considered. For other atoms, however, it is of considerable importance as first pointed out by Saika and Slichter¹⁴ for fluorine. The third term gives the effect of currents on neighboring atoms. At this level of approximation, it only involves the anisotropies of the local diamagnetic susceptibilities on other atoms. It is identical with the neighbor anisotropy formula that has been discussed in connection with the theory of proton shifts.^{9,15} In the present treatment, all three terms have been deduced within a single theoretical framework. Equation (5.10) is clearly a suitable starting point for discussing correlations between chemical shifts and atomic contributions to the diamagnetic susceptibility.

The expression (5.9) for the screening constant has some similarities to the result of a treatment given recently by Karplus and Das.¹⁶ These authors do not use a basis of gauge-invariant orbitals, but they avoid difficulties of the near cancellation of large terms by neglecting all terms involving orbitals on atoms other than A in their final expression. This leads to the first two terms of (5.9), but with the important difference that the expression (4.10) for $(\chi_p^A)_{zz}$ is replaced by the single contribution $B=A$. In many molecules contributions to $(\chi_p^A)_{zz}$ from other atoms B are considerable and may not be neglected.

¹⁴ A. Saika and C. P. Slichter, J. Chem. Phys. **22**, 26 (1954).

¹⁵ H. M. McConnell, J. Chem. Phys. **27**, 226 (1957).

¹⁶ M. Karplus and T. P. Das, J. Chem. Phys. **34**, 1683 (1961).

¹³ W. Lamb, Phys. Rev. **60**, 817 (1941).